

Cramer's rule

Definition

For an $n \times n$ matrix A and any vector $b \in \mathbb{R}^n$, we define $\det(A)_i(b)$ as the determinant of the matrix A with the i^{th} column replaced by b .

Theorem (Cramer's rule)

If A is invertible then the solution of $Ax = b$ is given by

$$x_i = \frac{\det(A)_i(b)}{\det(A)} \text{ for } i \leq n.$$

Ex of Cramer's rule:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

Solve $Ax = b$.

$$\det(A) = 1 \cdot \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ = 2.$$

- ① A is invertible and so
- ② we can use Cramer's rule.

$$\det(A)_1(b) = \det \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}$$

$$= \det \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$\uparrow$
 $C_1 - C_2$

$$= 1 \cdot \det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1$$

$$\det(A)_2(b) = \det \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 2.$$

$$\begin{aligned}
 \det(A)_3(b) &= \det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{pmatrix} \\
 &= \det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \\
 &\quad \quad \quad \uparrow \\
 &\quad \quad \quad C3 - C2 \\
 &= 3
 \end{aligned}$$

$$x_1 = \frac{1}{2}, \quad x_2 = \frac{2}{2} = 1, \quad x_3 = \frac{3}{2}.$$

Check: $\begin{pmatrix} \frac{1}{2} \\ 1 \\ \frac{3}{2} \end{pmatrix}$ is the solution to $Ax=b$.

- Not the guy from Seinfeld
- Gabriel Cramer, 1704 - 1752
- Mathematician, Université de Genève
- Offered a solution to the St. Petersburg Paradox besides computing solutions to systems of linear equations
- <https://mathshistory.st-andrews.ac.uk/Biographies/Cramer/>

The inverse via Cramer's rule

Definition

For an $n \times n$ matrix A , define the *adjugate* by

$$\text{adj}(A) = (C_{ij})^T,$$

the transpose of the matrix with $(i,j)^{\text{th}}$ entry C_{ij} .

Theorem (Explicit formula for the inverse)

If A is invertible then

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

Key point: The determinant of

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \text{ as a function}$$

of a_{ij} 's is a polynomial in n^2 variables.

Any cofactor C_{ij} is itself a determinant calculation possible multiplied by -1 ($C_{ij} = (-1)^{i+j} \det(A_{ij})$)

Again, each C_{ij} is a poly. in the inputs.

Ex of Finding the inverse using the adj.

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \quad \& \quad \det(A) = 2$$

$$C_{11} = 1$$

$$C_{31} = -1$$

$$C_{12} = (-1) \det \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix} = 0$$

$$C_{32} = 0$$

$$C_{13} = 1 \quad C_{13} = (-1)^{1+3} \det \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$$

$$C_{33} = 1$$

$$C_{21} = 1$$

$$= 1$$

$$C_{22} = 2$$

$$\boxed{C_{23} = -1}$$

$$C_{23} = (-1)^{2+3} \det \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = -1.$$

Cofactor matrix is $\begin{pmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 1 \end{pmatrix}$

$$\text{adj}(A) = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix} \cdot \quad \underline{\text{Check!}}$$

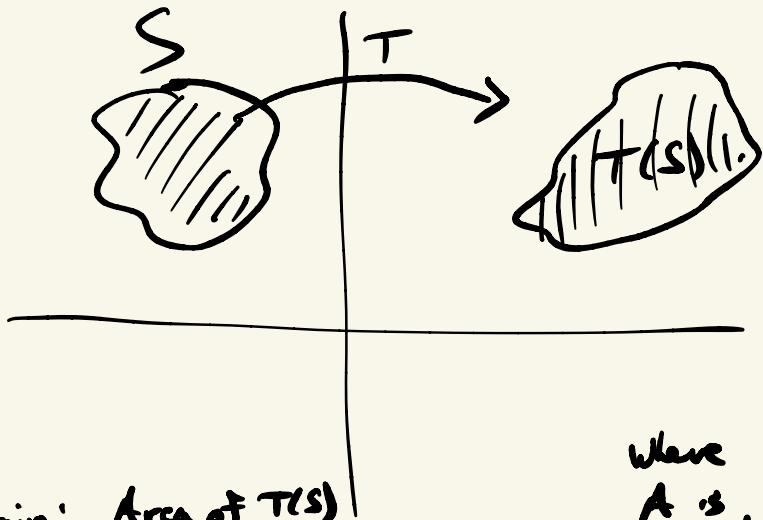
Area and Volume

Recall

- If A is the 2×2 matrix with columns v_1 and v_2 then the area of the parallelogram with sides v_1 and v_2 is $|\det(A)|$.
- If A is the 3×3 matrix with columns v_1 , v_2 and v_3 then the volume of the parallelepiped with sides v_1 , v_2 and v_3 is $|\det(A)|$.

Theorem

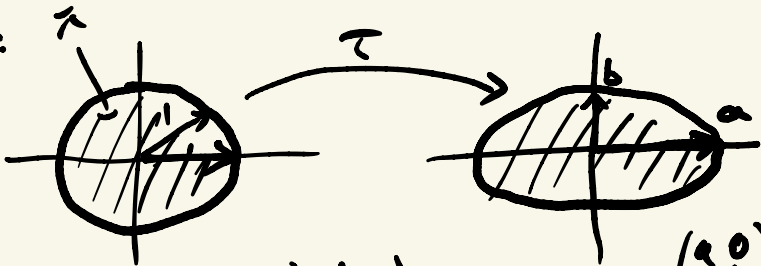
- 1 Suppose that S is a region in $\mathbb{R}^2$ with finite area and $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation with standard matrix A . Then the area of $T(S)$ is $|\det(A)| \times$ the area of S .
- 2 Suppose that S is a region in $\mathbb{R}^3$ with finite volume and $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear transformation with standard matrix A . Then the volume of $T(S)$ is $|\det(A)| \times$ the volume of S .



Claim: Area of $T(S)$
 $= (\text{Area of } S) \cdot |\det(A)|$

where
 A is
 the std.
 matrix for T .

Ex:



$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\det \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

Claim: The area of the ellipse is $\pi \cdot \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix}$
 $= ab\pi.$